AI-DND

HYBRID_TIME_SYSTEM.md

Hybrid World Time Simulation

Version 0.1

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1. DESIGN PHILOSOPHY

Time never stops.

However, time does not always flow at the same speed.

The world exists in three simulation states:

- Active Time
- Background Time
- Stone Sleep Time

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2. ACTIVE TIME

Active Time occurs while the player is online.

Simulation quality:

MAXIMUM.

The world simulates:

- NPC behavior;
- combat;
- dialogue;
- economy;
- weather;
- rumors;
- diseases;
- local events.

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3. BACKGROUND TIME

Background Time occurs when the player is online but far away.

Simulation quality:

MEDIUM.

The world simulates:

- economics;
- travel;
- wars;
- diseases;
- migration;
- politics.

Details are abstracted.

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4. STONE SLEEP TIME

Stone Sleep Time occurs when the player is offline.

Simulation quality:

ADAPTIVE.

The world continues to evolve.

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5. TIME COMPRESSION

Real time and game time differ.

Examples:

1 real minute: 1 game minute.

1 real hour: 2 game hours.

1 real day: 1-3 game days.

1 real week: 3-10 game days.

1 real month: 2-6 game weeks.

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6. LOCAL TIME

Areas near the player simulate:

- full AI;
- full schedules;
- full interactions.

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7. DISTANT TIME

Areas far away simulate:

- statistics;
- probabilities;
- summaries.

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8. NPC SIMULATION

Nearby NPC:

full simulation.

Far NPC:

abstract simulation.

Example:

Instead of:

"NPC walked 1427 steps."

Store:

"NPC traveled from village to city."

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9. WAR SIMULATION

Wars simulate using:

- armies;
- resources;
- morale;
- logistics.

Individual soldiers are not simulated.

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10. ECONOMIC SIMULATION

Economy updates:

- prices;
- shortages;
- trade routes;
- production.

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11. DISEASE SIMULATION

Diseases continue spreading.

Factors:

- population;
- climate;
- trade;
- medicine.

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12. MONSTER SIMULATION

Monsters:

- migrate;
- reproduce;
- hunt;
- die.

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13. PLAYER ABSENCE

During Stone Sleep:

- the player cannot act;
- the world continues.

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14. PLAYER RETURN

Upon awakening:

the simulation produces:

- world changes;
- rumors;
- personal consequences.

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15. EVENT IMPORTANCE

Events have priority.

LOW PRIORITY

Examples:

- weather;
- trade.

Simulated statistically.

MEDIUM PRIORITY

Examples:

- crime;
- disease;
- politics.

Simulated with detail.

HIGH PRIORITY

Examples:

- wars;
- deaths;
- major quests.

Simulated fully.

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16. NPC DEATH

NPCs may die while the player sleeps.

Examples:

- age;
- disease;
- combat;
- accidents.

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17. PLAYER CONNECTIONS

Important NPCs receive:

high simulation priority.

Examples:

- friends;
- teachers;
- enemies;
- companions.

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18. MEMORY PRESERVATION

Important memories are never removed.

Examples:

- betrayal;
- love;
- war;
- trauma.

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19. QUEST SIMULATION

Quests progress naturally.

Example:

Rescue quest.

Player absent:

Victim may die.

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20. FAMILY SIMULATION

Families continue living.

Examples:

- marriage;
- children;
- death;
- migration.

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21. CITY SIMULATION

Cities continue:

- building;
- trading;
- fighting;
- recovering.

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22. WORLD EVENTS

Events continue:

- invasions;
- epidemics;
- rebellions;
- disasters.

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23. HISTORY CREATION

Historical records continue.

Examples:

"The famine of year 842."

"The plague of Ashford."

"The disappearance of King Edric."

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24. PLAYER LEGEND

The player's legend changes while absent.

Rumors continue spreading.

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25. PERFORMANCE SYSTEM

Simulation complexity adjusts dynamically.

Near player: 100%.

Far away: 10%.

Offline: adaptive.

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26. WORLD CONSISTENCY

The world must remain believable.

Example:

A city cannot disappear overnight without cause.

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27. AI-DM RESPONSIBILITY

AI-DM determines:

- event significance;
- simulation priority;
- consequence generation.

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28. STONE SLEEP LIMITS

Stone Sleep cannot:

- destroy the game world;
- create impossible events;
- invalidate player progress.

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29. CORE QUESTION

When simulating the world, always ask:

"What would logically happen?"

Never ask:

"What would be convenient?"

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30. CORE RULE

The player pauses.

The world never does.